6.1.1

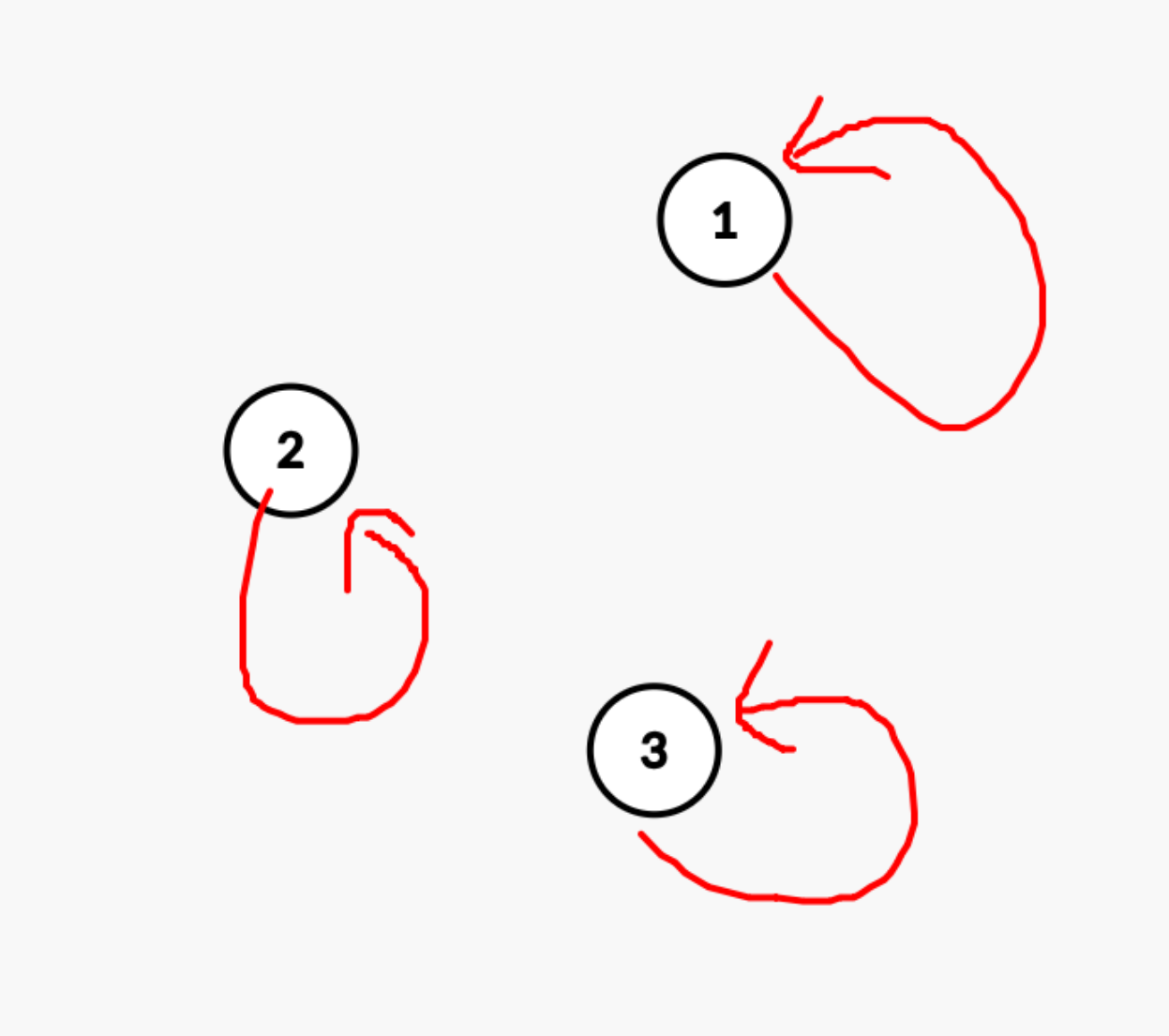


Figure 1: Pasted image 20261102172647.png

(c)

6.1.2

(e)

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Pairs:

$$(1,1), (1,4), (2,2), (3,3), (4,1), (4,4)$$

6.2.1

(c)

1. reflective
    1. For any positive integer x, we can choose $n = 1$ (which is a positive integer), and we get $x^1 = x$. Therefore, xPx holds for all x in the domain.
2. anti-symmetric
    1. $2P4$ because $2^2 = 4$, but $4P2$ is false because there is no positive integer n such that $4^n = 2$
3. transitive
    1. Suppose xPy and yPz. Then there exist positive integers n and m such that $y = x^n$ and $z = y^m$. Substituting the first equation into the second gives $z = (x^n)^m = x^{nm}$. Since n and m are positive integers, their product nm is also a positive integer. Thus, there exists a positive integer $k = nm$ such that $z = x^k$, which means xPz.

(i)
1. neither reflexive or anti-reflexive
1. For xTx to hold, we need $x + x = 0$, which simplifies to $2x = 0$ or $x = 0$. This is only true for $x = 0$, not for all real numbers
2. For $x = 0$, we have $0 + 0 = 0$, so $0T0$ is true.
2. symmetric
1. If xTy, then $x + y = 0$. By the commutative property of addition, $y + x = 0$, which means yTx.
3. not transitive
1. Consider $x = 1$, $y = -1$, and $z = 1$. We have xTy (since $1 + (-1) = 0$) and yTz (since $-1 + 1 = 0$). However, xTz is false because $1 + 1 = 2neq0$.

6.2.2

(a)

1. Reflexive
    1. $(m(x) gem(x))$ for every player (x).
2. Anti-symmetric
    1. If (xRy) and (yRx), then $(m(x) gem(y))$ and $(m(y) gem(x))$, so $(m(x) = m(y))$. Since no two distinct players have the same measured weight, this forces $(x = y)$.
3. Transitive
    1. if $(m(x) gem(y))$ and $(m(y) gem(z))$, then $(m(x) gem(z))$.

(b)

1. Reflexive
    1. $(m(x) gem(x))$ for every player (x).
2. Neither symmetric or anti-symmetric
    1. $(m(x) > m(y))$, we have (xRy) but not (yRx).
    2. There are distinct players $(xney)$ with $(m(x) = m(y))$, so (xRy) and (yRx)
3. Transitive
    1. if $(m(x) gem(y))$ and $(m(y) gem(z))$, then $(m(x) gem(z))$.

6.2.4

(c)

1. anti-reflexive
    1. $(a, a), (b, b), (c, c), (d, d) \notin R$.
2. symmetric
    1. every pair in R has its reverse in R.
3. not transitive
    1. $(a, b), (b, a) \in R$ but $(a, a) \notin R$.

6.3.3

(a) Largest in-degree vertex: (e); in-degree: (3).

(b) Largest out-degree vertex: (a); out-degree: (4).

(c) Self-loops: $(\langle a, a\rangle, \langle b, b\rangle, \langle e, e\rangle)$.

(d) Yes, $(\langle b, d, e, e\rangle)$ is a walk. It is an open walk because it starts at (b) and ends at (e).

(e) No, $(\langle a, c, f, g\rangle)$ is not a walk because there is no directed edge $(ctof)$.

(f) Yes, $(\langle a, a, c, e, e\rangle)$ is a walk. It is an open walk, and it is a trail but not a path, since vertices are repeated. So it is an open trail.

(g) One example is $(\langle b, b, d, e, e\rangle)$. It has length (4), is open, is a trail, and is not a path because (b) and (e) are repeated.

6.3.4

(a) A directed trail of length 5 exists:
$3 \to 1 \to 2 \to 2 \to 3 \to 2$,
using all five edges exactly once.

For every smaller length we also have examples:

$x = 0$: the trivial trail at any vertex
$x = 1$: $3 \to 1$
$x = 2$: $3 \to 1 \to 2$
$x = 3$: $3 \to 1 \to 2 \to 2$
$x = 4$: $3 \to 1 \to 2 \to 2 \to 3$
$x = 5$: $3 \to 1 \to 2 \to 2 \to 3 \to 2$

Therefore the possible trail lengths are
$x \in \{0, 1, 2, 3, 4, 5\}$.

6.4.2

(a) For real numbers x, y and z,

$$(x > y) \wedge (y \leq z)$$

Thus

$$\mathbb{R} \times \mathbb{R}$$

(b) For real numbers x, y and z,

$$(x \leq y) \wedge (y < z)$$

Thus

$$R_3$$

6.4.5

(a) *True.*
Let x be arbitrary.
Since R is reflexive, $(x, x) \in R$
Since S is reflexive, $(x, x) \in S$.
Taking $y = x$, we have xRy and ySx, so $(x, x) \in S \circ R$.
Hence $S \circ R$ is reflexive.

(b) *False.*
A counter is that let $A = a, b$, $R = (a, a), (a, b), (b, b)$ and $S = (a, a), (b, a), (b, b)$.
Both R and S are antisymmetric.
$(a, b) \in S \circ R$ because aRb and bSb
$(b, a) \in S \circ R$ because bRb and bSa
Since $a \neq b$, $S \circ R$ is not antisymmetric.

(c) *False.*
A counter is that let $A = a, b, c, u, v$, $R = (a, u), (b, v)$ and $S = (u, b), (v, c)$.
Both R and S are transitive vacuously, since neither has composable pairs.
$(a, b) \in S \circ R$ because aRu and uSb
$(b, c) \in S \circ R$ because bRv and vSc
But $(a, c) \notin S \circ R$, since the only R-image of a is u, and uSc is false. Thus $S \circ R$ is not transitive.

6.5.3

(b)

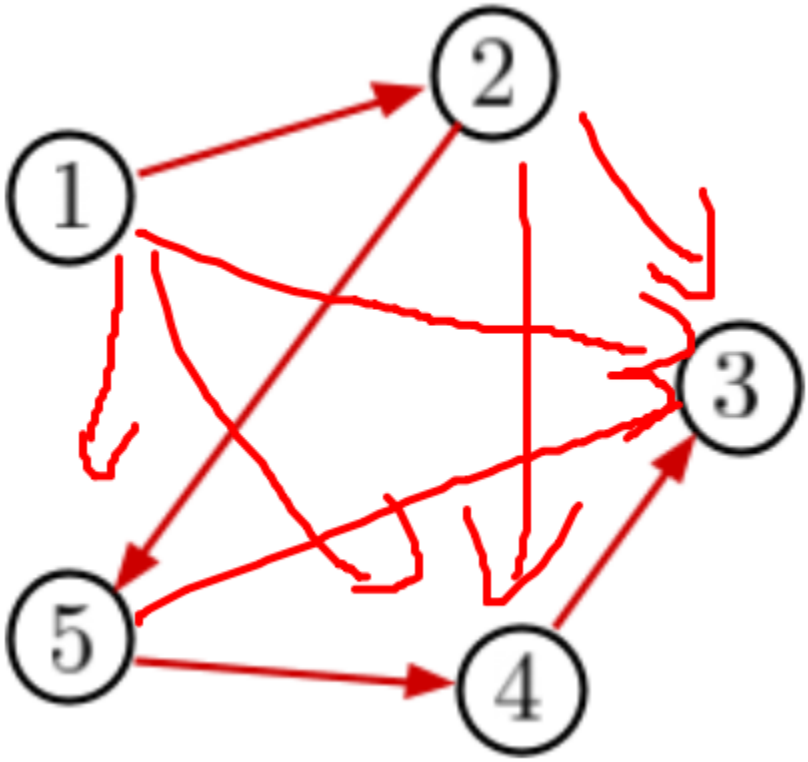


Figure 2: Pasted image 20261104112630.png

6.5.4

(a) no

(b) no

(c) yes

(d) no

(e) no

(f) yes

(g) vertices 2 and 4

(h) vertices 1 and 5

(i) yes, its out-degree in G+G+ is 5